

# Chahat Raj

## PhD Candidate in Computer Science | George Mason University

🌐 [chahatraj.github.io](https://chahatraj.github.io) @ [craj@gmu.edu](mailto:craj@gmu.edu) 📄 Google Scholar 📁 [github.com/chahatraj](https://github.com/chahatraj) 🐦 [chahatsaidit](https://twitter.com/chahatsaidit)

### Research Interests

My research focuses on NLP, particularly the evaluation and alignment of large language models. I study socio-cultural biases and other harms in multilingual and multimodal systems and develop methods to mitigate them, advancing trust-worthy and responsible AI. My work has been published at ACL, EMNLP, AIES, AAAI, and other venues.

### Education

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|----------------------|---------------------------------------------------------------------------------------------------------------------|--------------|
| May 2027<br>Aug 2022 | George Mason University<br>Ph.D. in Computer Science<br>Advisors: <b>Ziwei Zhu</b> , <b>Antonios Anastasopoulos</b> | Fairfax, VA  |
| Dec 2025<br>Aug 2022 | George Mason University<br>Masters in Computer Science                                                              | Fairfax, VA  |
| Aug 2021<br>Aug 2019 | Delhi Technological University<br>Masters in Information Systems (Thesis)<br>Advisor: <b>Priyanka Meel</b>          | Delhi, India |
| May 2019<br>Aug 2015 | Indira Gandhi Delhi Technical University for Women<br>Bachelors in Computer Science & Engineering                   | Delhi, India |

### Experience

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| Aug 2026<br>May 2026 | Amazon<br>Applied Scientist Intern   Mentor: <b>Ren Pang</b> , Manager: <b>Zheng Qi</b><br>> <i>A parameter-efficient safety steering technique for bedrock guardrails foundation models.</i>                                                                           | Seattle, WA |
| Aug 2025<br>May 2024 | George Mason University, University of Washington<br>Graduate Research Assistant   Advisors: <b>Ziwei Zhu</b> , <b>Antonios Anastasopoulos</b> , <b>Aylin Caliskan</b><br>> <i>Societal simulations, model unlearning, and cognitive alignment for bias mitigation.</i> | Fairfax, VA |
| Sep 2024<br>May 2024 | University of Washington   Tech Policy Lab, The Information School<br>Visiting Scholar   Advisor: <b>Aylin Caliskan</b><br>> <i>Bias, cognition, and mitigation in vision-language models.</i>                                                                          | Seattle, WA |
| Aug 2023<br>May 2023 | George Mason University   School of Computer Science<br>Graduate Research Assistant   Advisor: <b>Ziwei Zhu</b><br>> <i>Fairness, interpretability, and robustness in transformer-based fake news detection.</i>                                                        | Fairfax, VA |
| Jul 2022<br>Jun 2021 | University of Technology Sydney   School of Computer Science<br>Visiting Scholar   Advisor: <b>Mukesh Prasad</b><br>> <i>Cyberbullying, AI for Aboriginals, modeling disaster fundraising, and ML for clinical diagnosis.</i>                                           | Sydney, AU  |

### Selected Publications

[full list here](#)

- [C.12] **VIGNETTE: Socially Grounded Bias Evaluation for Vision-Language Models** [PDF | Code]  
**Chahat Raj**, Bowen Wei, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu  
*In Proceedings of the Association for Computational Linguistics: ACL '26, San Diego, California, USA* [ACL Main '26]
- [C.11] **Talent or Luck? Evaluating Attribution Bias in Large Language Models** [PDF | Code]  
**Chahat Raj**, Mahika Banerjee, Jinhao Pan, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu  
*In Findings of the Association for Computational Linguistics: ACL '26, San Diego, California, USA* [ACL Findings '26]
- [C.10] **Knowing Bias, Doing Better: Mitigating Social Bias in LLMs via Know-Bias Neuron Enhancement** [PDF | Code]  
Jinhao, **Chahat Raj**, Anjishnu Mukherjee, Sina Mansouri, Bowen Wei, Shloka Yada, Ziwei Zhu  
*International Conference in Machine Learning: ICML '26, Seoul, South Korea* [ICML '26]
- [C.9] **Measuring South Asian Biases in Large Language Models** [PDF | Code]  
Mamnuya Rinki, **Chahat Raj**, Anjishnu Mukherjee, Ziwei Zhu  
*TrustNLP at ACL '26* [TrustNLP '26]
- [C.8] **Discovering Bias Associations through Open-Ended LLM Generations** [PDF | Code]  
Jinhao Pan, **Chahat Raj**, Ziwei Zhu  
*In Proceedings of the AAAI Conference on Artificial Intelligence* [AAAI '26]

- [C.7] **What's Not Said Still Hurts: A Description-Based Evaluation Framework for Measuring Social Bias in LLMs**  
Jinhao Pan, [Chahat Raj](#), Ziyu Yao, Ziwei Zhu [[PDF](#) | [Code](#)]  
*In Findings of the Association for Computational Linguistics: EMNLP '25, Suzhou, China* [EMNLP Findings '25]
- [C.6] **Toward Inclusive Language Models: Sparsity-Driven Calibration for Systematic and Interpretable Mitigation of Social Biases in LLMs** [[PDF](#) | [Code](#)]  
Prommy Sultana Hossain, [Chahat Raj](#), Ziwei Zhu, Jessica Lin, Emanuela Marasco  
*In Findings of the Association for Computational Linguistics: EMNLP '25, Suzhou, China* [EMNLP Findings '25]
- [C.5] **BiasDora: Exploring Hidden Biased Associations in Vision-Language Models** [[PDF](#) | [Code](#)]  
[Chahat Raj](#), Anjishnu Mukherjee, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu  
*In Findings of the Association for Computational Linguistics: EMNLP '24, Miami, Florida, USA* [EMNLP Findings '24]
- [C.4] **Breaking Bias, Building Bridges: Evaluation and Mitigation of Social Biases in LLMs via Contact Hypothesis**  
[Chahat Raj](#), Anjishnu Mukherjee, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu [[PDF](#) | [Code](#)]  
*Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* [AIIES '24]
- [C.3] **Global Voices, Local Biases: Socio-cultural Prejudices across Languages** [[PDF](#) | [Code](#)]  
[Chahat Raj](#)\*, Anjishnu Mukherjee\*, Ziwei Zhu, Antonios Anastasopoulos  
*In Proceedings of EMNLP '23, Singapore* [EMNLP Main '23]
- [C.2] **SALSA: Salience-Based Switching Attack for Adversarial Perturbations in Fake News Detection Models**  
[Chahat Raj](#)\*, Anjishnu Mukherjee\*, Hemant Purohit, Antonios Anastasopoulos, Ziwei Zhu [[PDF](#) | [Code](#)]  
*European Conference on Information Retrieval* [ECIR '24]
- [C.1] **True and Fair: Robust and Unbiased Fake News Detection via Interpretable Machine Learning** [[PDF](#) | [Code](#)]  
[Chahat Raj](#), Anjishnu Mukherjee, Ziwei Zhu  
*Proceedings of the AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society* [AIIES '23]

#### Under Review

- [S.2] **Debias It Yourself: Teaching LLMs Cognitive Bias-Mitigation Interventions**  
[Chahat Raj](#), Sina Mansouri, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu  
[In Submission]
- [S.1] **Inoculation Prompting: Implicit Bias Mitigation in Large Language Models**  
Yongxu Sun, [Chahat Raj](#), Aylin Caliskan  
[In Submission]

## Selected Projects

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|-------------------------------------------|------------------------------------------------------------------------|----------------------------|
| > <b>AI, Power &amp; Society</b>          | Auditing Diffusion Models for Generational and Representational Errors | [ <a href="#">Report</a> ] |
| > <b>Computer Vision</b>                  | Cultural Image Translation for Human Figures                           | [ <a href="#">Report</a> ] |
| > <b>AI: Ethics, Policy &amp; Society</b> | Implicit Association Test in Multimodal Generative Models              | [ <a href="#">Report</a> ] |

## Teaching Assistantship

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|--------------------------------------------------------------|--------------------------------------------|----------------------|
| > <b>CS 484 Theory/Application Data Mining, GMU</b>          | Instr.: Keren Zhou                         | Spring '26           |
| > <b>CS 584 Theory/Application Data Mining, GMU</b>          | Instr.: Ziwei Zhu                          | Fall '25             |
| > <b>CS 108 Introduction to Computer Programming, GMU</b>    | Instr.: Sara Lynn Gopalakrishna, Bella Yin | Spring '24           |
| > <b>CS 478 Natural Language Processing, GMU</b>             | Instr.: Antonios Anastasopoulos            | Fall '23             |
| > <b>COMP 502 Mathematical Foundations of Computing, GMU</b> | Instr.: William D Ellis                    | Fall '23             |
| > <b>CS 112 Introduction to Python Programming, GMU</b>      | Instr.: Shahnaz Kamberi                    | Fall '22, Spring '23 |

## Student Mentorship

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|---------------------------------------------------|----------------------------------------------------------------------|
| > <b>Daksha Rajagopal [TJHSST → Purdue CS]</b>    | Modeling normative ethics and cross-cultural moral reasoning         |
| > <b>Angela Liu [TJHSST → Bryn Mawr College]</b>  | Assessing moral attributional biases in ambiguous contexts in LLMs   |
| > <b>Mahika Banerjee [TJHSST → KU Leuven]</b>     | Measuring attribution bias, coauthor on the ACL Findings '26 paper   |
| > <b>Mamnuya Rinki [MS Thesis, GMU → IntraFi]</b> | Mitigating South Asian Biases, coauthor on TrustNLP @ ACL '26 paper  |
| > <b>Yongxu Sun [MS, UW → PhD, UW]</b>            | Inoculation Prompting by in-context learning to debias LLM decisions |
| > <b>Diwita Banerjee [MS, GMU → DoorDash]</b>     | Assessing cuisine biases; scientific figure hallucination detection  |
| > <b>Sina Mansouri [MS, GMU]</b>                  | Long Context Memory Information Retrieval in LLM Agents              |
| > <b>Sharanya Garika [CS 678 Advanced NLP]</b>    | Identifying intersectional bias in LLMs across Indo-Aryan languages  |
| > <b>Aksh Patel [CS 678 Advanced NLP]</b>         | Identifying intersectional bias in LLMs across Indo-Aryan languages  |
| > <b>Samruddhi Deshmukh [CS 795 AI Ethics]</b>    | Auditing Text-to-Image Generation Models for Social Representation   |
| > <b>Abhi Krishna [CS 795 AI Ethics]</b>          | Auditing Text-to-Image Generation Models for Social Representation   |

## Awards & Grants

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| <ul style="list-style-type: none"><li>&gt; SAC Highlight Award at ACL, '26</li><li>&gt; Outstanding Academic Award at GMU, '26</li><li>&gt; Graduate Teaching Award at GMU, '26</li><li>&gt; Outstanding Reviewer Award at EACL '26</li><li>&gt; CAHMP Fellowship at GMU, '24</li><li>&gt; Best Paper Award at MASC-SLL '24</li><li>&gt; Summer Graduate Research Award at GMU, '23</li><li>&gt; OpenAI Researcher Access Program Award '25</li><li>&gt; Travel Grant by AAAI/ACM AIES '24 &amp; '23</li><li>&gt; Research Excellence Award at DTU '23 &amp; '22</li><li>&gt; Graduate Scholarship at DTU '19</li></ul> | <p>Vignette selected as top 15% of accepted papers.</p> <p>Received an Outstanding Academic Achievement Award.</p> <p>Received an Outstanding Graduate Teaching Assistant Award.</p> <p>Received an Outstanding Reviewer Award at EACL '26.</p> <p>Summer research on social biases in question answering.</p> <p>A Psychological View to Social Bias in LLMs.</p> <p>Explainability and Robustness of Debiasing.</p> <p>Funded USD 1,000 worth of OpenAI API credits.</p> <p>Funded USD 2,000 to attend AIES in San Jose and Montreal.</p> <p>Received three Commendable Research Awards with INR 150,000.</p> <p>Received INR 297,000 by AICTE for qualifying GATE.</p> |
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## Talks

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| <ul style="list-style-type: none"><li>&gt; VIGNETTE</li><li>&gt; LLM Ethics: Bias and Mitigation</li><li>&gt; LLM Ethics: Bias and Mitigation</li><li>&gt; Breaking Bias, Building Bridges</li><li>&gt; Breaking Bias, Building Bridges</li><li>&gt; Breaking Bias, Building Bridges</li><li>&gt; True and Fair</li><li>&gt; Cyberbully Monitoring</li></ul> | <p><i>Talk at ACL '26</i></p> <p><i>Invited Lecture, CS 678 Advanced NLP, Fall '24</i></p> <p><i>Invited Lecture, CS 782 Advanced ML, Spring '24</i></p> <p><i>Talk at AAAI/ACM AIES '24</i></p> <p><i>Talk at SouthNLP '24</i></p> <p><i>Talk at MASC-SLL '24</i></p> <p><i>Talk at AAAI/ACM AIES '23</i></p> <p><i>ML4AU Research Showcase '21</i></p> | <p>San Diego, CA</p> <p>George Mason, VA</p> <p>George Mason, VA</p> <p>San Jose, CA</p> <p>Emory, GA</p> <p>Johns Hopkins, MD</p> <p>Montreal, QC</p> <p>Remote</p> |
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## Program Committee

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- > **Reviewer** NeurIPS '26, ARR '23-'26, CHI '26, AAAI '26, IASEAI '26, COLM '25-'26, TrustNLP '24-'26, LTEDI @ EACL '24, AIES '23-'26, CSCW '24, NeurIPS '23, ACM TIST